Can a religious perspective lead to a different work ethic? How?

Wikipedia Entries

Christianity

Christianity is an Abrahamic monotheistic religion, which states that Jesus is the Son of God and rose from the dead after his crucifixion, whose coming as the messiah (Christ) was prophesied in the Old Testament and chronicled in the New Testament. It is the world's largest and most widespread religion with over 2.3 billion followers, comprising around 28.8% of the world population. Its adherents, known as Christians, are estimated to make up a majority of the population in 120 countries and territories.

Christianity remains culturally diverse in its Western and Eastern branches, and doctrinally diverse concerning justification and the nature of salvation, ecclesiology, ordination, and Christology. Most Christian denominations, however, generally hold in common the belief that Jesus is God the Son—the Logos incarnated—who ministered, suffered, and died on a cross, but rose from the dead for the salvation of humankind; this message is called the gospel, meaning the "good news". The four canonical gospels of Matthew, Mark, Luke and John describe the life of Jesus and teachings as preserved in the early Christian tradition, with the Old Testament as the gospels' respected background.

Christianity began in the 1st century, after the death of Jesus, as a Judaic sect with Hellenistic influence in the Roman province of Judaea. The disciples of Jesus spread their faith around the Eastern Mediterranean area, despite significant persecution. The inclusion of Gentiles led Christianity to slowly separate from Judaism in the 2nd century. Emperor Constantine I decriminalized Christianity in the Roman Empire in 313 AD with the Edict of Milan. In 325 AD,

Constantine convened the Council of Nicaea, where the Nicene Creed was adopted, and where Early Christianity was consolidated into what would become the state religion by around 380 AD. The Church of the East and Oriental Orthodoxy both split over differences in Christology during the 5th century, while the Eastern Orthodox Church and the Catholic Church separated in the East–West Schism in 1054. Protestantism split into numerous denominations from the Catholic Church during the Reformation era (16th century). Following the Age of Discovery (15th–17th century), Christianity expanded globally via missionary work, evangelism, immigration, and trade. Christianity played a prominent role in the development of Western civilization, particularly in Europe from late antiquity and the Middle Ages.

The Reformation

The Reformation, also known as the Protestant Reformation or the European Reformation,was a time of major theological movement in Western Christianity in 16th-century Europe that posed a religious and political challenge to the papacy and the authority of the Catholic Church hierarchy. Towards the end of the Renaissance, the Reformation marked the beginning of Protestantism. It is considered one of the events that signified the end of the Middle Ages and the beginning of the early modern period in Europe.

The Reformation is usually dated from Martin Luther's publication of the Ninety-five Theses in 1517, which gave birth to Lutheranism. Prior to Martin Luther and other Protestant Reformers, there were earlier reform movements within Western Christianity. The end of the Reformation era is disputed among modern scholars.

Max Weber

Maximilian Carl Emil Weber was a German sociologist, historian, jurist, and political economist who was one of the central figures in the development of sociology and the social sciences more generally. His ideas continue to influence social theory and research.

Born in Erfurt in 1864, Weber studied law and history in Berlin, Göttingen, and Heidelberg. After earning his doctorate in law in 1889 and habilitation in 1891, he taught in Berlin, Freiburg, and Heidelberg. He married his cousin Marianne Schnitger two years later. In 1897, he had a breakdown after his father died following an argument. Weber ceased teaching and travelled until the early 1900s. He recovered and wrote The Protestant Ethic and the Spirit of Capitalism. During the First World War, he initially supported Germany's war effort but became critical of it and supported democratisation. He also gave the lectures "Science as a Vocation" and "Politics as a Vocation". After the war, Weber co-founded the German Democratic Party, unsuccessfully ran for office, and advised the drafting of the Weimar Constitution. Meanwhile, he resumed teaching in

Vienna and Munich. He died of pneumonia in 1920 at the age of 56, possibly as a result of the post-war Spanish flu pandemic. A book, Economy and Society, was left unfinished.

Protestant Work Ethic

The Protestant work ethic, also known as the Calvinist work ethic or the Puritan work ethic, is a work ethic concept in the social sciences, particularly sociology, economics, and history. It emphasizes that a person's subscription to the values espoused by the Protestant faith, particularly Calvinism, results in diligence, discipline, and frugality. The term was initially coined by the sociologist Max Weber in his 1905 book, The Protestant Ethic and the Spirit of Capitalism. In this work, Weber asserted that Protestant ethics and values, along with the Calvinist doctrines of asceticism and predestination, enabled the rise and spread of capitalism. Just as priests and caring professionals are deemed to have a vocation (or "calling" from God) for their work, according to the Protestant work ethic, the "lowly" workman also has a noble vocation which he can fulfill through dedication to his work.

A Short Note on Reading the Bible

Benjamin Parry, Proceedings of the PRG, April 2026

The Bible is a small library of books. Bringing together ancient Jewish and early Christian writings. A strange collection of myths, histories, stories, songs, poems, proverbs, prophecies. Yet despite this vast diversity Christians read the book as a coherent collection of ideas that are interrelated and describe the ways that God has revealed himself to humanity.

Usually Christians will treat reading the Bible as an ongoing practice. Rather than reading the book all the way through they will follow reading plans individually or with their church to cover the major arcs of the story over the course of a year (sometimes as part of what is called the liturgical calendar). Or, they turn to specific poems or passages for guidance.

As such ‘reading the Bible’ can be a fraught exercise and it is easy to misunderstand the context for sections without a map of the book. So on the next two pages there is a rough map of the different sections of the Bible.

The sections we have chosen follow. These represent some of the foundational ideas in the myths, laws, wisdom literature and early Christian stories about the nature of work.

Genesis 1–3
Deuteronomy 5:12–15
Ecclesiastes 1–3
Luke 10:38–42
Matthew 6:25–34

The Old Testament

The Law (~700 BC — 500BC)
Origins myths for the world and the early Jewish community as well as the first commandments

Histories (~700 BC — 300 BC)
History of the ancient state of Israel including the reign of King David, his son Solomon, and the destruction of the first Jewish state

Wisdom Lit (~900 BC — 300 BC)
Poems & proverbs on the right way to live often attributed to David and Solomon.

Prophets (~800 BC - 150 BC)
Jewish prophecies about the future. Sometimes considered some of the earliest predictions in world history. Many concerning a messiah and future restoration of Israel. Christians later read as prophecies concerning Jesus Christ. "Christ" is the Greek word for messiah.

Genesis	Gn	1
Exodus	Ex	47
Leviticus	Lv	85
Numbers	Nm	112
Deuteronomy	Dt	151
Joshua	Jos	184
Judges	Jdg	207
Ruth	Ru	229
1 Samuel	1Sm	233
2 Samuel	2Sm	262
1 Kings	1Kg	287
2 Kings	2Kg	317
1 Chronicles	1Ch	345
2 Chronicles	2Ch	374
Ezra	Ezr	407
Nehemiah	Neh	418
Esther	Est	433
Job	Jb	440
Psalms	Ps	472
Proverbs	Pr	554
Ecclesiastes	Ec	586
Song of Songs	Sg	594
Isaiah	Is	600
Jeremiah	Jr	664
Lamentations	Lm	726
Ezekiel	Ezk	734
Daniel	Dn	782
Hosea	Hs	797
Joel	Jl	807
Amos	Am	811
Obadiah	Ob	819
Jonah	Jnh	821
Micah	Mc	823
Nahum	Nah	829
Habakkuk	Hab	832
Zephaniah	Zph	835
Haggai	Hg	839
Zechariah	Zch	841
Malachi	Mal	849

The New Testament

The Gospels (60-90 AD)
Accounts of the life of Jesus including stories and sayings

Early Church History (90 AD)

Letters (50 - 100 AD)
Letters between the earliest Christian communities explaining their beliefs and giving recommendations to each other. Most written by the apostle Paul.

Apocalyptic Prophecy (90 AD)

Matthew	Mt	855
Mark	Mk	887
Luke	Lk	907
John	Jn	941
Acts	Ac	966
Romans	Rm	997
1 Corinthians	1Co	1011
2 Corinthians	2Co	1023
Galatians	Gl	1031
Ephesians	Eph	1036
Philippians	Php	1040
Colossians	Col	1043
1 Thessalonians	1Th	1046
2 Thessalonians	2Th	1049
1 Timothy	1Tm	1051
2 Timothy	2Tm	1055
Titus	Ti	1058
Philemon	Phm	1060
Hebrews	Heb	1061
James	Jms	1071
1 Peter	1Pt	1075
2 Peter	2Pt	1079
1 John	1Jn	1082
2 John	2Jn	1086
3 John	3Jn	1087
Jude	Jd	1088
Revelation	Rv	1089

Genesis 1

The Creation of the World

1 In the beginning, God created the heavens and the earth. **2** The earth was without form and void, and darkness was over the face of the deep. And the Spirit of God was hovering over the face of the waters.

3 And God said, "Let there be light," and there was light. **4** And God saw that the light was good. And God separated the light from the darkness. **5** God called the light Day, and the darkness he called Night. And there was evening and there was morning, the first day.

6 And God said, "Let there be an expanse in the midst of the waters, and let it separate the waters from the waters." **7** And God made the expanse and separated the waters that were under the expanse from the waters that were above the expanse. And it was so. **8** And God called the expanse Heaven. And there was evening and there was morning, the second day.

9 And God said, "Let the waters under the heavens be gathered together into one place, and let the dry land appear." And it was so. **10** God called the dry land Earth, and the waters that were gathered together he called Seas. And God saw that it was good.

11 And God said, "Let the earth sprout vegetation, plants yielding seed, and fruit trees bearing fruit in which is their seed, each according to its kind, on the earth." And it was so. **12** The earth brought forth vegetation, plants yielding seed according to their own kinds, and trees bearing fruit in which is their seed, each according to its kind. And God saw that it was good. **13** And there was evening and there was morning, the third day.

14 And God said, "Let there be lights in the expanse of the heavens to separate the day from the night. And let them be for signs and for seasons, and for days and years, **15** and let them be lights in the expanse of the heavens to give light upon the earth." And it was so. **16** And God made the two great lights—the greater light to rule the day and the lesser light to rule the night—and the stars. **17** And God set them in the expanse of the heavens to give light on the earth, **18** to rule over the day and over the night, and to separate the light from the darkness. And God saw that it was good. **19** And there was evening and there was morning, the fourth day.

20 And God said, "Let the waters swarm with swarms of living creatures, and let birds fly above the earth across the expanse of the heavens." **21** So God created the great sea creatures and every living creature that moves, with which the waters swarm, according to their kinds, and every winged bird according to its kind. And God saw that it was good. **22** And God blessed them, saying, "Be fruitful and multiply and fill the waters in the seas, and let birds multiply on the earth." **23** And there was evening and there was morning, the fifth day.

24 And God said, "Let the earth bring forth living creatures according to their kinds—livestock and creeping things and beasts of the earth according to their kinds." And it was so. **25** And God

made the beasts of the earth according to their kinds and the livestock according to their kinds, and everything that creeps on the ground according to its kind. And God saw that it was good.

26 Then God said, "Let us make man in our image, after our likeness. And let them have dominion over the fish of the sea and over the birds of the heavens and over the livestock and over all the earth and over every creeping thing that creeps on the earth."

27 So God created man in his own image, in the image of God he created him; male and female he created them.

28 And God blessed them. And God said to them, "Be fruitful and multiply and fill the earth and subdue it, and have dominion over the fish of the sea and over the birds of the heavens and over every living thing that moves on the earth." **29** And God said, "Behold, I have given you every plant yielding seed that is on the face of all the earth, and every tree with seed in its fruit. You shall have them for food. **30** And to every beast of the earth and to every bird of the heavens and to everything that creeps on the earth, everything that has the breath of life, I have given every green plant for food." And it was so. **31** And God saw everything that he had made, and behold, it was very good. And there was evening and there was morning, the sixth day.

Genesis 2

The Seventh Day, God Rests

1 Thus the heavens and the earth were finished, and all the host of them. **2** And on the seventh day God finished his work that he had done, and he rested on the seventh day from all his work that he had done. **3** So God blessed the seventh day and made it holy, because on it God rested from all his work that he had done in creation.

The Creation of Man and Woman

4 These are the generations of the heavens and the earth when they were created, in the day that the Lord God made the earth and the heavens.

5 When no bush of the field was yet in the land and no small plant of the field had yet sprung up—for the Lord God had not caused it to rain on the land, and there was no man to work the ground, **6** and a mist was going up from the land and was watering the whole face of the ground— **7** then the Lord God formed the man of dust from the ground and breathed into his nostrils the breath of life, and the man became a living creature. **8** And the Lord God planted a garden in Eden, in the east, and there he put the man whom he had formed. **9** And out of the ground the Lord God made to spring up every tree that is pleasant to the sight and good for food. The tree of life was in the midst of the garden, and the tree of the knowledge of good and evil.

10 A river flowed out of Eden to water the garden, and there it divided and became four rivers. **11** The name of the first is the Pishon. It is the one that flowed around the whole land of Havilah, where there is gold. **12** And the gold of that land is good; bdellium and onyx stone are there. **13** The name of the second river is the Gihon. It is the one that flowed around the whole land of Cush. **14** And the name of the third river is the Tigris, which flows east of Assyria. And the fourth river is the Euphrates.

15 The Lord God took the man and put him in the garden of Eden to work it and keep it. **16** And the Lord God commanded the man, saying, "You may surely eat of every tree of the garden, **17** but of the tree of the knowledge of good and evil you shall not eat, for in the day that you eat of it you shall surely die."

18 Then the Lord God said, "It is not good that the man should be alone; I will make him a helper fit for him." **19** Now out of the ground the Lord God had formed every beast of the field and every bird of the heavens and brought them to the man to see what he would call them. And whatever the man called every living creature, that was its name. **20** The man gave names to all livestock and to the birds of the heavens and to every beast of the field. But for Adam there was not found a helper fit for him. **21** So the Lord God caused a deep sleep to fall upon the man, and while he slept took one of his ribs and closed up its place with flesh. **22** And the rib that the Lord God had taken from the man he made into a woman and brought her to the man. **23** Then the man said,

"This at last is bone of my bones and flesh of my flesh; she shall be called Woman, because she was taken out of Man."

24 Therefore a man shall leave his father and his mother and hold fast to his wife, and they shall become one flesh. **25** And the man and his wife were both naked and were not ashamed.

Genesis 3

The Fall

1 Now the serpent was more crafty than any other beast of the field that the Lord God had made.

He said to the woman, "Did God actually say, 'You shall not eat of any tree in the garden'?" **2** And the woman said to the serpent, "We may eat of the fruit of the trees in the garden, **3** but God said, 'You shall not eat of the fruit of the tree that is in the midst of the garden, neither shall you touch it, lest you die.'" **4** But the serpent said to the woman, "You will not surely die. **5** For God knows that when you eat of it your eyes will be opened, and you will be like God, knowing good and evil." **6** So when the woman saw that the tree was good for food, and that it was a delight to the eyes, and that the tree was to be desired to make one wise, she took of its fruit

and ate, and she also gave some to her husband who was with her, and he ate. **7** Then the eyes of both were opened, and they knew that they were naked. And they sewed fig leaves together and made themselves loincloths.

8 And they heard the sound of the Lord God walking in the garden in the cool of the day, and the man and his wife hid themselves from the presence of the Lord God among the trees of the garden. **9** But the Lord God called to the man and said to him, "Where are you?" **10** And he said, "I heard the sound of you in the garden, and I was afraid, because I was naked, and I hid myself." **11** He said, "Who told you that you were naked? Have you eaten of the tree of which I commanded you not to eat?" **12** The man said, "The woman whom you gave to be with me, she gave me fruit of the tree, and I ate." **13** Then the Lord God said to the woman, "What is this that you have done?" The woman said, "The serpent deceived me, and I ate."

14 The Lord God said to the serpent,

"Because you have done this, cursed are you above all livestock and above all beasts of the field; on your belly you shall go, and dust you shall eat all the days of your life. **15** I will put enmity between you and the woman, and between your offspring and her offspring; he shall bruise your head, and you shall bruise his heel."

16 To the woman he said,

"I will surely multiply your pain in childbearing; in pain you shall bring forth children. Your desire shall be for your husband, and he shall rule over you."

17 And to Adam he said,

"Because you have listened to the voice of your wife and have eaten of the tree of which I commanded you, 'You shall not eat of it,' cursed is the ground because of you; in pain you shall eat of it all the days of your life; **18** thorns and thistles it shall bring forth for you; and you shall eat the plants of the field. **19** By the sweat of your face you shall eat bread, till you return to the ground, for out of it you were taken; for you are dust, and to dust you shall return."

20 The man called his wife's name Eve, because she was the mother of all living. **21** And the Lord God made for Adam and for his wife garments of skins and clothed them.

22 Then the Lord God said, "Behold, the man has become like one of us in knowing good and evil. Now, lest he reach out his hand and take also of the tree of life and eat, and live forever—" **23** therefore the Lord God sent him out from the garden of Eden to work the ground from which he was taken. **24** He drove out the man, and at the east of the garden of Eden he placed the cherubim and a flaming sword that turned every way to guard the way to the tree of life.

Deuteronomy 5:12–15

12 "'Observe the Sabbath day, to keep it holy, as the Lord your God commanded you. **13** Six days you shall labor and do all your work, **14** but the seventh day is a Sabbath to the Lord your God. On it you shall not do any work, you or your son or your daughter or your male servant or your female servant, or your ox or your donkey or any of your livestock, or the sojourner who is within your gates, that your male servant and your female servant may rest as well as you. **15** You shall remember that you were a slave in the land of Egypt, and the Lord your God brought you out from there with a mighty hand and an outstretched arm. Therefore the Lord your God commanded you to keep the Sabbath day.

Ecclesiastes 1

All Is Vanity

1 The words of the Preacher, the son of David, king in Jerusalem.

2 Vanity of vanities, says the Preacher, vanity of vanities! All is vanity. **3** What does man gain by all the toil at which he toils under the sun? **4** A generation goes, and a generation comes, but the earth remains forever. **5** The sun rises, and the sun goes down, and hastens to the place where it rises. **6** The wind blows to the south and goes around to the north; around and around goes the wind, and on its circuits the wind returns. **7** All streams run to the sea, but the sea is not full; to the place where the streams flow, there they flow again. **8** All things are full of weariness; a man cannot utter it; the eye is not satisfied with seeing, nor the ear filled with hearing. **9** What has been is what will be, and what has been done is what will be done, and there is nothing new under the sun. **10** Is there a thing of which it is said, "See, this is new"? It has been already in the ages before us. **11** There is no remembrance of former things, nor will there be any remembrance of later things yet to be among those who come after.

Ecclesiastes 2

The Vanity of Toil

18 I hated all my toil in which I toil under the sun, seeing that I must leave it to the man who will come after me, **19** and who knows whether he will be wise or a fool? Yet he will be master of all for which I toiled and used my wisdom under the sun. This also is vanity. **20** So I turned about and gave my heart up to despair over all the toil of my labors under the sun, **21** because sometimes a person who has toiled with wisdom and knowledge and skill must leave everything to be enjoyed by someone who did not toil for it. This also is vanity and a great evil. **22** What has a man from all the toil and striving of heart with which he toils beneath the sun? **23** For all

his days are full of sorrow, and his work is a vexation. Even in the night his heart does not rest. This also is vanity.

24 There is nothing better for a person than that he should eat and drink and find enjoyment in his toil. This also, I saw, is from the hand of God, **25** for apart from him who can eat or who can have enjoyment? **26** For to the one who pleases him God has given wisdom and knowledge and joy, but to the sinner he has given the business of gathering and collecting, only to give to one who pleases God. This also is vanity and a striving after wind.

Ecclesiastes 3

A Time for Everything

1 For everything there is a season, and a time for every matter under heaven:

2 a time to be born, and a time to die; a time to plant, and a time to pluck up what is planted; **3** a time to kill, and a time to heal; a time to break down, and a time to build up; **4** a time to weep, and a time to laugh; a time to mourn, and a time to dance; **5** a time to cast away stones, and a time to gather stones together; a time to embrace, and a time to refrain from embracing; **6** a time to seek, and a time to lose; a time to keep, and a time to cast away; **7** a time to tear, and a time to sew; a time to keep silence, and a time to speak; **8** a time to love, and a time to hate; a time for war, and a time for peace.

The God-Given Task

9 What gain has the worker from his toil? **10** I have seen the business that God has given to the children of man to be busy with. **11** He has made everything beautiful in its time. Also, he has put eternity into man's heart, yet so that he cannot find out what God has done from the beginning to the end. **12** I perceived that there is nothing better for them than to be joyful and to do good as long as they live; **13** also that everyone should eat and drink and take pleasure in all his toil—this is God's gift to man.

14 I perceived that whatever God does endures forever; nothing can be added to it, nor anything taken from it. God has done it, so that people fear before him. **15** That which is, already has been; that which is to be, already has been; and God seeks what has been driven away.

From Dust to Dust

16 Moreover, I saw under the sun that in the place of justice, even there was wickedness, and in the place of righteousness, even there was wickedness. **17** I said in my heart, God will judge the righteous and the wicked, for there is a time for every matter and for every work. **18** I said in my

heart with regard to the children of man that God is testing them that they may see that they themselves are but beasts. **19** For what happens to the children of man and what happens to the beasts is the same; as one dies, so dies the other. They all have the same breath, and man has no advantage over the beasts, for all is vanity. **20** All go to one place. All are from the dust, and to dust all return. **21** Who knows whether the spirit of man goes upward and the spirit of the beast goes down into the earth? **22** So I saw that there is nothing better than that a man should rejoice in his work, for that is his lot. Who can bring him to see what will be after him?

Luke 10:38–42

Martha and Mary

38 Now as they went on their way, Jesus entered a village. And a woman named Martha welcomed him into her house. **39** And she had a sister called Mary, who sat at the Lord's feet and listened to his teaching. **40** But Martha was distracted with much serving. And she went up to him and said, "Lord, do you not care that my sister has left me to serve alone? Tell her then to help me." **41** But the Lord answered her, "Martha, Martha, you are anxious and troubled about many things, **42** but one thing is necessary. Mary has chosen the good portion, which will not be taken away from her."

Matthew 6:25–34

Do Not Be Anxious

25 "Therefore I tell you, do not be anxious about your life, what you will eat or what you will drink, nor about your body, what you will put on. Is not life more than food, and the body more than clothing? **26** Look at the birds of the air: they neither sow nor reap nor gather into barns, and yet your heavenly Father feeds them. Are you not of more value than they? **27** And which of you by being anxious can add a single hour to his span of life? **28** And why are you anxious about clothing? Consider the lilies of the field, how they grow: they neither toil nor spin, **29** yet I tell you, even Solomon in all his glory was not arrayed like one of these. **30** But if God so clothes the grass of the field, which today is alive and tomorrow is thrown into the oven, will he not much more clothe you, O you of little faith? **31** Therefore do not be anxious, saying, 'What shall we eat?' or 'What shall we drink?' or 'What shall we wear?' **32** For the Gentiles seek after all these things, and your heavenly Father knows that you need them all. **33** But seek first the kingdom of God and his righteousness, and all these things will be added to you.

34 "Therefore do not be anxious about tomorrow, for tomorrow will be anxious for itself. Sufficient for the day is its own trouble.

Max Weber, The Protestant Ethic and the Spirit of Capitalism. 1905

Chapter III Luther's Conception of the Calling

Now it is unmistakable that even in the German word Beruf, and perhaps still more clearly in the English calling, a religious conception, that of a task set by God, is at least suggested. The more emphasis is put upon the word in a concrete case, the more evident is the connotation. And if we trace the history of the word through the civilized languages, it appears that neither the predominantly Catholic peoples nor those of classical antiquity have possessed any expression of similar connotation for what we know as a calling (in the sense of a life-task, a definite field in which to work), while one has existed for all predominantly Protestant peoples. It may be further shown that this is not due to any ethnical peculiarity of the languages concerned. It is not, for instance, the product of a Germanic spirit, but in its modern meaning the word comes from the Bible translations, through the spirit of the translator, not that of the original. In Luther's translation of the Bible it appears to have first been used at a point in Jesus Sirach (xi. 20 and 21) precisely in our modern sense. After that it speedily took on its present meaning in the everyday speech of all Protestant peoples, while earlier not even a suggestion of such a meaning could be found in the secular literature of any of them, and even, in religious writings, so far as I can ascertain, it is only found in one of the German mystics whose influence on Luther is well known.

Like the meaning of the word, the idea is new, a product of the Reformation. This may be assumed as generally known. It is true that certain suggestions of the positive valuation of routine activity in the world, which is contained in this conception of the calling, had already existed in the Middle Ages, and even in late Hellenistic antiquity. We shall speak of that later. But at least one thing was unquestionably new: the valuation of the fulfillment of duty in worldly affairs as the highest form which the moral activity of the individual could assume. This it was which inevitably gave everyday worldly activity a religious significance, and which first created the conception of a calling in this sense. The conception of the calling thus brings out that central dogma of all Protestant denominations which the Catholic division of ethical precepts into preecepta and consilia discards. The only way of living acceptably to God was not to surpass worldly morality in monastic asceticism, but solely through the fulfillment of the obligations imposed upon the individual by his position in the world. That was his calling.

Luther developed the conception in the course of the first decade of his activity as a reformer. At first, quite in harmony with the prevailing tradition of the Middle Ages, as represented, for example, by Thomas Aquinas he thought of activity in the world as a thing of the flesh, even though willed by God. It is the indispensable natural condition of a life of faith, but in itself, like eating and drinking, morally neutral. But with the development of the conception of sola fide in all its consequences, and its logical result, the increasingly sharp emphasis against the Catholic consilia evangelica of the monks as dictates of the devil, the calling grew in importance. The

monastic life is not only quite devoid of value as a means of justification before God, but he also looks upon its renunciation of the duties of this world as the product of selfishness, withdrawing from temporal obligations. In contrast, labor in a calling appears to him as the outward expression of brotherly love. This he proves by the observation that the division of labor forces every individual to work for others, but his viewpoint is highly naive, forming an almost grotesque contrast to Adam Smith's well known statements on the same subject. However, this justification, which is evidently essentially scholastic, soon disappears again, and there remains, more and more strongly emphasized, the statement that the fulfillment of worldly duties is under all circumstances the only way to live acceptably to God. It and it alone is the will of God, and hence every legitimate calling has exactly the same worth in the sight of God.

That this moral justification of worldly activity was one of the most important results of the Reformation, especially of Luther's part in it, is beyond doubt, and may even be considered a platitude. This attitude is worlds removed from the deep hatred of Pascal, in his contemplative moods, for all worldly activity, which he was deeply convinced could only be understood in terms of vanity or low cunning. And it differs even more from the liberal utilitarian compromise with the world at which the Jesuits arrived. But just what the practical significance of this achievement of Protestantism was in detail is dimly felt rather than clearly perceived.

In the first place it is hardly necessary to point out that Luther cannot be claimed for the spirit of capitalism in the sense in which we have used that term above, or for that matter in any sense whatever. The religious circles which today most enthusiastically celebrate that great achievement of the Reformation are by no means friendly to capitalism in any sense. And Luther himself would, without doubt, have sharply repudiated any connection with a point of view like that of Franklin. Of course, one cannot consider his complaints against the great merchants of his time, such as the Fuggers, as evidence in this case. For the struggle against the privileged position, legal or actual, of single great trading companies in the sixteenth and seventeenth centuries may best be compared with the modern campaign against the trusts, and can no more justly be considered in itself an expression of a traditionalistic point of view. Against these people, against the Lombards, the monopolists, speculators, and bankers patronized by the Anglican Church and the kings and parliaments of England and France, both the Puritans and the Huguenots carried on a bitter struggle. Cromwell, after the battle of Dunbar (September 1650), wrote to the Long Parliament: "Be pleased to reform the abuses of all professions: and if there be any one that makes many poor to make a few rich, that suits not a Commonwealth." But, nevertheless, we will find Cromwell following a quite specifically capitalistic line of thought . On the other hand, Luther's numerous statements against usury or interest in any form reveal a conception of the nature of capitalistic acquisition which, compared with that of late Scholasticism, is, from a capitalistic viewpoint, definitely backward. Especially, of course , the doctrine of the sterility of money which Anthony of Florence had already refuted.

But it is unnecessary to go into detail. For, above all the consequences of the conception of the calling in the religious sense for worldly conduct were susceptible to quite different interpretations. The effect of the Reformation as such was only that, as compared with the Catholic attitude, the moral emphasis on and the religious sanction of, organized worldly labor in

a calling was mightily increased. The way in which the concept of the calling, which expressed this change, should develop further depended upon the religious evolution which now took place in the different Protestant Churches. The authority of the Bible, from which Luther thought he had derived his idea of the calling, on the whole favored a traditionalistic interpretation. The old Testament, in particular, though in the genuine prophets it showed no sign of a tendency to excel worldly morality, and elsewhere only in quite isolated rudiments and suggestions, contained a similar religious idea entirely in this traditionalistic sense. Everyone should abide by his living and let the godless run after gain. That is the sense of all the statements which bear directly on worldly activities. Not until the Talmud is a partially, but not even then fundamentally, different attitude to be found. The personal attitude of Jesus is characterized in classical purity by the typical antique Oriental plea: “Give us this day our daily bread.” The element of radical repudiation of the world, as expressed in the (mamwnaς thς adikiaς), excluded the possibility that the modern idea of calling should be based on his personal authority. In the apostolic era as expressed in the New Testament, especially in St. Paul, the Christian looked upon worldly activity either with indifference, or at least essentially traditionalistically; for those first generations were filled with eschatological hopes. Since everyone was simply waiting for the coming of the Lord, there was nothing to do but remain in the station and in the worldly occupation in which the call of the Lord had found him, and labor as before. Thus he would not burden his brothers as an object of charity, and it would only be for a little while. Luther read the Bible through the spectacles of his whole attitude; at the time and in the course of his development from about 1518 to 1530 this not only remained traditionalistic but became ever more so.

In the first years of his activity as a reformer he was, since he thought of the calling as primarily of the flesh, dominated by an attitude closely related, in so far as the form of world activity was concerned, to the Pauline eschatological indifference as expressed in I Cor. vii. One may attain salvation in any walk of life; on the short pilgrimage of life there is no use in laying weight on the form of occupation. The pursuit of material gain beyond personal needs must thus appear as a symptom of lack of grace, and since it can apparently only be attained at the expense of others, directly reprehensible. As he became increasingly involved in the affairs of the world, he came to value work in the world more highly. But in the concrete calling an individual pursued he saw more and more a special command of God to fulfill these particular duties which the Divine Will had imposed upon him. And after the conflict with the Fanatics and the peasant disturbances, the objective historical order of things in which the individual has been placed by God becomes for Luther more and more a direct manifestation of divine will. The stronger and stronger emphasis on the providential element, even in particular events of life, led more and more to a traditionalistic interpretation based on the idea of Providence. The individual should remain once and for all in the station and calling in which God had placed him, and should restrain his worldly activity within the limits imposed by his established station in life. While his economic traditionalism was originally the result of Pauline indifference, it later became that of a more and more intense belief in divine providence, which identified absolute obedience to God’s will, with absolute acceptance of things as they were. Starting from this background, it was impossible for Luther to establish a new or in any way fundamental connection between worldly activity and religious principles. His acceptance of purity of doctrine as the one infallible criterion of the

Church, which became more and more irrevocable after the struggles of the twenties, was in itself sufficient to check the development of new points of view in ethical matters.

Thus for Luther the concept of the calling remained traditionalistic. His calling is something which man has to accept as a divine ordinance, to which he must adapt himself. This aspect outweighed the other idea which was also present, that work in the calling was a, or rather the, task set by God. And in its further development, orthodox Lutheranism emphasized this aspect still more. Thus, for the time being, the only ethical result was negative; worldly duties were no longer subordinated to ascetic ones; obedience to authority and the acceptance of things as they were, were preached. In this Lutheran form the idea of a calling had, as will be shown in our discussion of medieval religious ethics, to a considerable extent been anticipated by the German mystics. Especially in Tauler's equalization of the values of religious and worldly occupations, and the decline in valuation of the traditional forms of ascetic practices on account of the decisive significance of the ecstatic-contemplative absorption of the divine spirit by the soul. To a certain extent Lutheranism means a step backward from the mystics, in so far as Luther, and still more his Church, had, as compared with the mystics, partly undermined the psychological foundations for a rational ethics. (The mystic attitude on this point is reminiscent partly of the Pietest and partly of the Quaker psychology of faith.) That was precisely because he could not but suspect the tendency to ascetic self discipline of leading to salvation by works, and hence he and his Church were forced to keep it more and more in the background.

Thus the mere idea of the calling in the Lutheran sense is at best of questionable importance for the problems in which we are interested. This was all that was meant to be determined here. But this is not in the least to say that even the Lutheran form of the renewal of the religious life may not have had some practical significance for the objects of our investigation; quite the contrary. Only that significance evidently cannot be derived directly from the attitude of Luther and his Church to worldly activity, and is perhaps not altogether so easily grasped as the connection with other branches of Protestantism. It is thus well for us next to look into those forms in which a relation between practical life and a religious motivation can be more easily perceived than in Lutheranism. We have already called attention to the conspicuous part played by Calvinism and the Protestant sects in the history of capitalistic development. As Luther found a different spirit at work in Zwingli than in himself, so did his spiritual successors in Calvinism. And Catholicism has to the present day looked upon Calvinism as its real opponent.

Now that may be partly explained on purely political grounds. Although the Reformation is unthinkable without Luther's own personal religious development, and was spiritually long influenced by his personality, without Calvinism his work could not have had permanent concrete success. Nevertheless, the reason for this common repugnance of Catholics and Lutherans lies, at least partly, in the ethical peculiarities of Calvinism. A purely superficial glance shows that there is here quite a different relationship between the religious life and earthly activity than in either Catholicism or Lutheranism. Even in literature motivated purely by religious factors that is evident. Take for instance the end of the Divine Comedy, where the poet in Paradise stands speechless in his passive contemplation of the secrets of God, and compare it

with the poem which has come to be called the Divine Comedy of Puritanism. Milton closes the last song of Paradise Lost after describing the expulsion from paradise as follows: -

“They, looking back, all the eastern side beheld
Of paradise, so late their happy seat,
Waved over by that flaming brand; the gate
With dreadful faces thronged and fiery arms.
Some natural tears they dropped, but wiped them soon:
The world was all before them, there to choose
Their place of rest, and Providence their guide.”

And only a little before Michael had said to Adam:

“ ... Only add
Deeds to thy knowledge answerable; add faith;
Add virtue, patience, temperance; add love,
By name to come called Charity, the soul
Of all the rest: then wilt thou not be loth
To leave this Paradise, but shall possess
A Paradise within thee, happier far.”

One feels at once that this powerful expression of the Puritan’s serious attention to this world, his acceptance of his life in the world as a task, could not possibly have come from the pen of a medieval writer. But it is just as uncongenial to Lutheranism, as expressed for instance in Luther’s and Paul Gerhard’s chorales. It is now our task to replace this vague feeling by a somewhat more precise logical formulation, and to investigate the fundamental basis of these differences. The appeal to national character is generally a mere confession of ignorance, and in this case it is entirely untenable. To ascribe a unified national character to the Englishmen of the seventeenth century would be simply to falsify history. Cavaliers and Roundheads did not appeal to each other simply as two parties, but a radically distinct species of men, and whoever look into the matter carefully must agree with them. 0n the other hand, a difference of character between the English merchant adventurers and the old Hanseatic merchants is not to be found; nor can any other fundamental difference between the English and German characters at the end of the Middle Ages, which cannot easily be explained by the differences of their political history. It was the power of religious influence, not alone, but more than anything else, which created the differences of which we are conscious today.

We thus take as our starting point in the investigation of the relationship between the old Protestant ethic and the spirit of capitalism the works of Calvin, of Calvinism, and the other Puritan sects. But it is not to be understood that we expect to find any of the founders or representatives of these religious movements considering the promotion of what we have called the spirit of capitalism as in any sense the end of his life-work. We cannot well maintain that the pursuit of worldly goods, conceived as an end in itself, was to any of them of positive ethical value. Once and for all it must be remembered that programs of ethical reform never were at the

center of interest for any of the religious reformers (among whom, for our purposes, we must include men like Menno, George Fox, and Wesley). They were not the founders of societies for ethical culture nor the proponents of humanitarian projects for social reform or cultural ideals. The salvation of the soul and that alone was the center of their life and work. Their ethical ideals and the practical results of their doctrines were all based on that alone, and were the consequences of purely religious motives. We shall thus have to admit that the cultural consequences of the Reformation were to a great extent, perhaps in the particular aspects with which we are dealing predominantly, unforeseen and even unwished-for results of the labors of the reformers. They were often far removed from or even in contradiction to all that they themselves thought to attain.

The following study may thus perhaps in a modest way form a contribution to the understanding of the manner in which ideas become effective forces in history. In order, however, to avoid any misunderstanding of the sense in which any such effectiveness of purely ideal motives is claimed at all, I may perhaps be permitted a few remarks in conclusion to this introductory discussion.

In such a study, it may at once be definitely stated, no attempt is made to evaluate the ideas of the Reformation in any sense, whether it concern their social or their religious worth. We have continually to deal with aspects of the Reformation which must appear to the truly religious consciousness as incidental and even superficial. For we are merely attempting to clarify the part which religious forces have played in forming the developing web of our specifically worldly modern culture, in the complex interaction of innumerable different historical factors. We are thus inquiring only to what extent certain characteristic features of this culture can be imputed to the influence of the Reformation. At the same time we must free ourselves from the idea that it is possible to deduce the Reformation, as a historically necessary result, from certain economic changes. Countless historical circumstances, which cannot be reduced to any economic law, and are not susceptible of economic explanation of any sort, especially purely political processes, had to concur in order that the newly created Churches should survive at all.

On the other hand, however, we have no intention whatever of maintaining such a foolish and doctrinaire thesis as that the spirit of capitalism (in the provisional sense of the term explained above) could only have arisen as the result of certain effects of the Reformation, or even that capitalism as an economic system is a creation of the Reformation. In itself, the fact that certain important forms of capitalistic business organization are known to be considerably older than the Reformation is a sufficient refutation of such a claim. On the contrary, we only wish to ascertain whether and to what extent religious forces have taken part in qualitative formation and the quantitative expansion of that spirit over the world. Furthermore, what concrete aspects of our capitalistic culture can be traced to them, In view of the tremendous confusion of interdependent influences between the material basis, the forms of social and political organization, and the ideas current in the time of the Reformation, we can only proceed by investigating whether and at what points certain correlations between forms of religious belief and practical ethics can be worked out. At the same time we shall as far as possible clarify the manner and the general direction in which, by virtue of those relationships, the religious

movements have influenced the development of material culture. Only when this has been determined with reasonable accuracy can the attempt be made to estimate to what extent the historical development of modern culture can be attributed to those religious forces and to what extent to others.

Chapter 13

Puritanism and British Exceptionalism

The Industrial Revolution, as every schoolchild knows, started in Britain. The discussion so far has emphasized a set of European phenomena to explain the exceptionalism of Europe. Was Britain somehow unique even within Europe? I have summarized much of this literature elsewhere (Mokyr, 2009a), but given the importance of culture and beliefs, a natural question that could and should be raised is whether British culture was somehow different from the rest of Europe. The problem is that much of European intellectual culture was transnational. Newton, Descartes, Spinoza, Galileo, and Leibniz were read by intellectuals from Stockholm to Madrid and from Dublin to St. Petersburg. Where Britain may have differed is in the emergence of the Puritan movement. The historical importance of Puritanism to the growth of useful knowledge in England was placed on the agenda of historians by Robert K. Merton ([1938] 2001) in his classic work linking Puritanism to the rise of modern science. He stressed the importance of Puritanism to the particular form that British scientific culture developed in the seventeenth century. Merton's unit of analysis was very much a set of cultural beliefs and values, not a formal set of theological propositions. These were shared by many individuals who were not formally Puritans and in this sense his thesis has led to a somewhat pointless debate about the role of religion in the Scientific Revolution and to what extent he can be proven wrong by showing that many of the key scientists in this period were not Puritans.

The advance of Baconianism among the British elite in the seventeenth century was an integral part of a religious cultural movement. A series of scholars in the 1930s made this point with great emphasis. Richard Foster Jones, in a book first published in 1936, argued for a growing influence of Baconianism in Britain that eventually would lead to the Scientific Revolution. In a memorable phrase he wrote that "our modern scientific

utilitarianism is the offspring of Bacon begot upon Puritanism" (Jones, [1936] 1961, p. 88). Bacon's work was highly congenial to Puritan ideology, even though Bacon was no Puritan.

The term "sentiment," which Merton uses quite frequently, is much like the cultural elements I defined in chapter 1. He stressed that the seventeenth century was a profoundly religious age in which the concern about salvation permeated daily life to a degree that is quite unimaginable today, and that when religious beliefs were not in alignment with economic progress, one of the two had to give in (Merton, [1938] 2001, p. 91). The growth of Puritan culture in British society represents an example of choice-based social learning: people became committed to various versions of Puritanism, because they found its religious arguments compelling, its leaders inspiring, or because it was consonant with a set of prior beliefs. By itself, there was nothing unusual in the commitment of Puritans to scientific and technological progress. It was part of a much larger European movement in which intellectuals, including not just English Puritans but also many Italian Catholics and German Lutherans, found science consistent with their religious beliefs and indeed an attractive way to practice them. Religion guided and at times encouraged scientific progress, but it could also seriously constrain its agenda in unexpected ways. For instance, as we have seen, Jesuits were in many ways good and eager scientists, yet in the end the religious dogmas to which they were subjected imposed institutional constraints that doomed them as a progressive institution. Orthodox Calvinism, similarly, could be seriously problematic.

What is important is that Puritans were deeply attracted to experimental philosophy, which they found to be congruent with their beliefs. Cohen (2012, p. 574) has pointed out that experimental research was as remote an activity from sacrilege and atheism that one could find. Experimental scientists such as Boyle insisted that their investigations of nature would make no sense without a creator who in his wisdom designed a universe in which bodies obeyed rules that scientists themselves could not possibly understand and he explicitly invoked the clockmaker analogy, in which an intricate mechanism could not emerge without an intelligent designer (Boyle, 1664, pp. 71–72). As Shapin (1994, pp. 156 ff) has stressed, Boyle's Puritanism meant above all that his identity as a scholar was strongly flavored by his Christianity and that he fulfilled a holy mission by becoming "a priest of nature" and giving thanks and praises to his maker for all of creation. But his religiosity also served to enhance his trustworthiness in an age in which atheism was still equated by most people— including intellectuals—with amorality.

The importance of Merton's thesis is not that it explains the Scientific Revolution, which it does not. Puritanism was English, the Scientific Revolution was European. The growth of science in Europe required a transnational institution, such as the Republic of Letters. But it required more: the coevolution of culture and local institutions, a major example of

which was the coevolution of Puritanism and the English part of the Republic of Letters. The significance of Merton's interpretation is that it explicitly rejected a materialist approach to religion at this age. It is based on the assumption that people adopted certain ideological beliefs in large part because they found their content persuasive and compelling or for any of the other evolutionary biases; any facile materialist notion that links their cultural choices directly to their being members of a "bourgeoisie" or a commercial class is rightfully rejected.[1] Such leading intellectuals as Wilkins, Boyle, the botanists John Ray and Francis Willughby, the mathematician John Wallis, the physician and chemist Jonathan Goddard, and the political economist William Petty were all committed Puritans. It is not easy to associate Puritanism as such directly with any specific scientific advance, but Puritans greatly enhanced the social prestige of experimental science and thus helped prepare the ground for the Industrial Enlightenment. Their political preeminence was short-lived, but in restoration England most of them were received into the Royalist camp supported by the "protection of a lenient King" (Jones, [1936] 1961, p. 270). They abandoned their hopes of taking control of the universities and instead created their own organizations and environment in which they could practice the experimental Baconian science they believed in.

Puritans embraced science, in part because it simultaneously "manifested the Glory of God and enhanced the Good of Man" (Merton, 1973, p. 232). For them, as Webster ([1975] 2002, p. 505) has remarked, the ideal life was one that efficiently deployed one's ability for personal advantage and public service and glorified God by maximizing one's material resources. These two objectives were not separable but complemented each other in ways that took until the end of the seventeenth century to be fully worked out. Devout individuals recognized the profound ethical implications of scientific investigation: the systematic and meticulous study of God's creation was the closest a Calvinist could get to an inscrutable deity that could not be grasped by the cultivated intellect. The Puritan ideology built upon the Baconian belief that experimental science was a Christian religious activity; Puritanism and Science thus found a common ground in empiricism and experimentalism. Robert Boyle placed on the title page of his book *The Christian Virtuoso* (1690) the statement that the book would show "that by being addicted to Experimental Philosophy, a Man is being assisted rather than indisposed to being a Good Christian." The Puritans were the beneficiaries of what has been called the "Baconian compromise," which held that the key to God's works (that is, the understanding of the physical environment) is the key to God's word, so that the

[1] Indeed, it is strking that even though the political influence of the commercial-financial class was rising with the success of the Whigs, in its early stages the Royal Society had few representatives of this class, despite its strong interest in manufacturing techniques.

study of nature instructs interpreters of the scriptures. Science could be pursued free from the constraints of metaphysical systems and the literal interpretation of biblical statements about the natural world, as long as the book of nature could be interpreted as divine revelation (Moore, 1986, p. 323).

All this was paired to the utilitarian belief, inspired by Bacon, that the study of natural regularities and phenomena would lead to breakthroughs in useful knowledge that would eventually "refer their Attainment to the Glory of the Great Author of Nature and to the Comfort of Mankind," as Boyle wrote in his will (quoted in Merton, 1973, p. 235). Merton stressed that Puritanism was especially compatible with a labor-intensive approach to science (as experimentalism no doubt was), because it inherently condemned idleness as morally repulsive.[2] Others have objected that it was not just Puritans who adopted this viewpoint, and that not all Puritans saw eye to eye on this point. Yet Puritanism was a powerful cultural innovation in the sixteenth century and gathered strength in the decades following Bacon's death in 1626. Its success in the competitive marketplace for ideas in England deserves special attention precisely because it was part of a larger cultural movement in Western Europe that searched for ways to reconcile their religious beliefs, the pressure of organized religion, and their scientific interests. Puritanism itself was not necessarily the taproot of the technological flourishing of the eighteenth century, but it was indicative of something much deeper that characterized the Republic of Letters.

Moreover, Merton argued that seventeenth century Puritanism, whether it was loyal to the Church of England, Presbyterian, or vacillating between them, associated a great deal of virtue with "good works," which were increasingly associated with utilitarian objectives and useful knowledge. Merton pointed to another highly successful cultural entrepreneur, the Puritan theologian Richard Baxter (1615–1691), as the source of these beliefs. It may seem odd for an economic historian to point to Baxter as an important figure in the economic development of the West, but Baxter's influence on both sides of the Atlantic in the eighteenth century was considerable, and Max Weber regarded him as the author who "stands out above many other writers on Puritan ethics, both because of his eminently practical and realistic attitude, and, at the same time, because of the

[2] Bishop Sprat, himself no Puritan but clearly reflecting the views of some of the dominant Puritan members of the early Royal Society, confronted head on the charge raised against experimental science that it spent too much effort on studying nature and not enough on reflecting on salvation and the next world. He argued strongly that "true and unfain'd Mortification [asceticism sanctified by abstemiousness and contemplation] is not at all inconsistent with mens consultation of their happiness in this world or being emploi'd about earthly affairs. The honest pursuit of the conveniences, decence, and ornaments of a mortal condition ... is by no means contradictory to the most real and severe duties of a Christian" and that "the happiness and security of mankind in this life" were profound Christian values (Sprat, 1667, pp. 367–68).

universal recognition accorded to his works" (Weber, [1905] 1958, p. 155). His idea of the glorification of God through "good works" focused on labor that was "useful and profitable in a worldly sense ... at first blush, sheer utilitarianism" (Merton, [1938] 2001, p. 62).[3] Baxter was a cultural entrepreneur relying above all on content-based bias: he tried to persuade people by using reason and argument in the belief that everyone was amenable to reason. He was the author of an estimated 130 books and pamphlets; a modern site lists 67 titles. His "works enjoyed an unprecedented popularity, many going through repeated printings. Puritanism had always utilized the press, but there had never been a literary career like this, either in scale or in success: Baxter was the first author of a string of best-sellers in British literary history" (Keeble, 2004).

Religion in Britain, then, was not just a permissive factor; for many of the scientists in the second half of the seventeenth century, scientific writing and research was a mode of worship. That basic tenet of Robert Merton seems to have survived close scrutiny. It took the Enlightenment to shift the weight of emphasis from the glorification of God to the material well-being of humankind, but the seeds had been planted. As has been pointed out many times (for example, Shapin, 1988a), Merton did not argue strict causality in the sense that Puritanism was a sufficient or necessary condition for the Scientific Revolution and the Industrial Enlightenment. Much of the pathbreaking scientific work in seventeenth-century Europe was carried out on the Continent, a substantial amount of it by scientists active in Catholic countries. But Merton's argument that a growing accommodation between religious beliefs and scientific endeavor was of considerable significance in the rise of British science has survived the test of time. Equally important, because of its increasingly pragmatic and empirical orientation and interest in prescriptive knowledge, and its diffusion to a wide circle through mostly private and spontaneous mechanisms, eighteenth-century British science may have been a more suitable background to the Industrial Enlightenment. Members of such dissenting religions such as Unitarianists and Quakers have been widely observed to have played a role much larger than their relative numbers would indicate (Inkster, 1991, pp. 42–45; Jacob, 2000a).

Puritan science, if such a concept can be defined at all, was deeply empirical and shied away from the highly deductive, logically rigorous constructs, such as those favored by Descartes in France and Hobbes in Britain. It relied heavily on Baconian experimental methodology, on observation, and on a careful examination of facts rather than on a rational, carefully argued set of propositions that follow from one another. This can be

[3] Baxter at some point told Robert Boyle that he took great pleasure in experimental learning, and Boyle commented that Baxter was not one of those "narrow-souled Divines" who was suspicious of natural philosophy (Jones, [1936] 1961, p. 323).

seen especially by examining the work of the most important Puritan mathematician, John Wallis (1616–1703), the first mathematician to join the Royal Society and a Puritan clergyman who worked as hard on his sermons as he did on his mathematical proofs. Appointed the Savilian Professor of Mathematics at Oxford in 1649, most of his important work came afterward. His mathematics was in many ways flavored by an experimental, Baconian approach. As Alexander (2014, pp. 262–78) points out, Wallis's mathematics was pragmatic and experimental. It "replicated the experimental practices of the Royal Society ... relied on induction and never claimed to arrive at a final truth; the ultimate arbiter of truth was the consensus of men" (Alexander, 2014, pp. 277–78). Early in his career he had written that "speculative knowledge [that is, deductive arguments] is found even in the Devils in as large and ample measure as in the Saints." In contrast, "experimentall knowledge, is of another nature, whereby we know what we know. ...We not only Know that it is so but we Tast and See it to be so ... For Truths thus clearly and sensibly (as it were) reveiled to the Soul, it seems not to be in the power of the will to reject" (Wallis, 1643, pp. 60–61). Such an empirical position may seem a curious one to take for a mathematician, but in fact his *Arithmetica Infinitorum* (1656) with its firm belief in infinitesimals was important to young Newton in developing his own mathematics of fluxions (Westfall, 1980, pp. 113–17). It is clear from Newton's own writings, and those of many of his contemporaries, that Newton's mechanical system of the world reflected a rational and powerful Deity, and in that regard his work is consistent with the ideas of Puritanism.

The advantage of models of cultural evolution is that they are contingent and concern ex ante probabilities rather than deterministic causal models. In other words, they force us to recognize that things could have turned out differently than they did with fairly minor changes in initial conditions or accidents along the way. The Puritans and the early members of the Royal Society were one set of players in a large European arena. Their idea of doing science was very different from that of their many opponents, who were not nearly as committed to the inductive and empirical modes. In England, they battled one of the most powerful intellects of the Puritan age, Thomas Hobbes (who never was elected to the Society but had a highly influential position in Charles II's court) and on the Continent they contended with Cartesians and the Jesuits alike. Had political and military developments taken different turns, conservative forces might have prevailed and taken a more hostile attitude toward the new interpretation of the world. There was nothing inexorable in the ultimate triumph of the desirability of scientific progress any more than, say, the eventual evolution of *Homo sapiens* (or any other specific species) on the planet.

At the same time, contingency should not be overdone, and evolutionary theory, while making substantial allowances for contingency, does

IZA Discussion Paper No. 2886
June 2007

ABSTRACT

Was Weber Wrong?
A Human Capital Theory of Protestant Economic History*

Max Weber attributed the higher economic prosperity of Protestant regions to a Protestant work ethic. We provide an alternative theory, where Protestant economies prospered because instruction in reading the Bible generated the human capital crucial to economic prosperity. County-level data from late 19th-century Prussia reveal that Protestantism was indeed associated not only with higher economic prosperity, but also with better education. We find that Protestants' higher literacy can account for the whole gap in economic prosperity. Results hold when we exploit the initial concentric dispersion of the Reformation to use distance to Wittenberg as an instrument for Protestantism.

JEL Classification: N33, Z12, I20

Keywords: human capital, Protestantism, economic history

Corresponding author:

Ludger Woessmann
CESifo
Poschingerstr. 5
81679 Munich
Germany
E-mail: woessmann@ifo.de

* The idea that the rise of Protestantism may have fostered human capital accumulation in Europe stems from our discussions with Paul E. Peterson, who again traces the idea back to a late-1960s University of Chicago seminar by C. Arnold Anderson and Mary Jean Bowman. We have received substantive comments during seminar presentations at Ifo Institute Munich, Aarhus Business School, ZEW Mannheim, the theology and economics faculties of the University of Munich, the University of Zurich, the Third Christmas Conference of German Expatriate Economists, WZB Berlin, the Max Planck Institute for Research in Collective Goods in Bonn, and the Universities of Florence, Rome "Tor Vergata", Harvard, Stockholm, Stanford, and UC Davis. Discussions with and comments from Andreas Ammermüller, Knut Borchardt, David Card, Andreas Diekmann, Guido Friebel, Claudia Goldin, Avner Greif, Mathias Hoffmann, Tim Lorentzen, Volker Meier, Paul Peterson, Holger Sieg, and Ulrich Woitek were very fruitful. Support has come from the Program on Education Policy and Governance of Harvard University. Erik Hornung and Martin Hofmann provided capable research assistance. We are grateful to all of them.

I. Introduction

In the century since Max Weber [1904/05] (2001) suggested in *The Protestant Ethic and the Spirit of Capitalism* that a "Protestant ethic" was instrumental for economic progress, several interpretations have emerged how the greater economic affluence of Protestants relative to Catholics might have come about (if at all). Weber's seminal work is considered a key founding stone of sociology, and leading mechanisms emphasized in sociological interpretations include that Protestants' specific ethic may have induced them to work harder and to save more. To the sociological theories, we offer a simple alternative economic theory based on the standard human capital model. We suggest that Luther's demand that all Christians should be able to read the Gospel by themselves led to increased literacy among Protestants which, incidentally, could then also be used in economic activities.

In order to test empirically which of the suggested mechanisms may be right or wrong, we go on to present county-level evidence from the very perspective that Weber had, namely Weber's native Prussia as the main part of the German Empire in late 19th century. The evidence reveals that there is indeed a significant positive association between Protestantism and economic success. We then show that there is also a significant positive association between Protestantism and literacy. Intriguingly, after controlling for the positive effect of literacy on economic success, there remains no significant difference in economic success between Protestant and Catholic counties. Human capital can account for the whole denominational difference in economic affluence, leaving little scope for any denomination-based work ethic. Furthermore, we argue that the approximately concentric diffusion of Protestantism around Luther's city of Wittenberg in Lutheran times allows us to identify exogenous variation in Protestantism in late 19th century. We find that even when instrumented by distance to Wittenberg, a county's share of Protestants increases literacy, which in turn yields higher economic progressiveness. The results suggest that while religious affiliation may indeed have had economic consequences, this may have been for reasons unrelated to any ethical disposition.

It is well known that Luther opposed the Roman Catholic practice of reading out the Gospel in the scholarly language of Latin, and that he delivered the first influential translation of the Bible into his native German, for everybody to read. What is less well known today is that he also explicitly favored universal schooling, for the simple reason that people had to be literate in order to be able to read God's Word, the Bible. In light of human capital theory, the ensuing literacy has the unintended side effect in the economic realm to increase people's productivity and thus prosperity. In this view, religion in the form of Protestant denomination may well be associated with economic affluence, but not because of

any difference in work ethic, but rather incidentally because it furthered the creation of human capital. In the next section, we introduce this effect into a simple model of human capital.[1]

To study the relationship between Protestantism, education, and economic prosperity empirically, 19th century Prussia is the natural place to look at. The birthplace of Martin Luther is on 1871 Prussian territory, where the Reformation was initiated and where his doctrine diffused in its purest form. Prussia is also Max Weber's birthplace, and the situation in late 19th century is the one he had in mind when formulating his thesis. The core of Prussia is a well-defined region with rather uniform laws and institutional settings, so that empirical investigation is not hampered by institutional heterogeneity.[2] Prussia is also reasonably well divided between Protestants and Catholics, at roughly two thirds to one third of the population in 1871, so that no denomination constitutes just a small minority, and Prussia had a long tradition of freedom of religion. Finally, the Prussian Statistical Office collected an impressive amount of data, available at the level of 452 counties. It is generally accepted that Prussian orderliness and thoroughness yielded high-quality data even in the 19th century, which are available in archives to these days. Important for our purposes, the 1871 Prussian Census is the first instance surveying literacy of the whole population. We thus do not have to rely on data from selective samples like military recruits, which provide only a limited picture of the population at large.

The use of Prussian county data allows us to go beyond the existing empirical literature on the Weber thesis, which mostly uses cross-country variation. Iannaccone (1998) concludes that the empirical literature on the relationship between Protestantism and economic outcomes largely rejects the empirical validity of Weber's argument. In fact, studies based on cross-country data seem to show no relationship at all between Protestantism and the rise of industrial capitalism. Delacroix and Nielsen (2001) conclude that there is no clear-cut pattern in the adoption of capitalism across European countries in the 19th century with respect to the dominant religion.[3]

A broader context of papers studies the association between religion and economic outcomes. Quite generally, religion is an important expression of culture (Guiso, Sapienza, and Zingales 2006), and as such is viewed as a possible fundamental cause of economic growth. Thus, Barro and McCleary (2003; 2005) study the association between different religions and economic growth. In a study

---

[1] In a closely related argument, Botticini and Eckstein (2005, 2006) suggest a human capital interpretation of Jewish history, where the ultimate root of Jewish economic prosperity as merchants lies in a centuries-old Judaic rule that required male Jews to be able to read the Torah in the synagogue, and to teach the reading of the Torah to their sons.

[2] While this may not hold strictly for recent Prussian annexations, we will show below that all our results hold in an analysis restricted to the core Prussian territory which had been part of Prussia for decades and even centuries.

[3] In a growth model calibrated to England, Cavalcanti, Parente, and Zhao (2006) suggest that Protestantism could at best account for only slight delays in the start of industrialization.

concerned with controlling the effects of economic institutions, Acemoglu, Johnson, and Robinson (2001, p. 1388; 2005, p. 419) find no effect of religion on growth in a cross-country setting.

Any cross-country study is plagued by the difficulty of disentangling the effect of religion from other possible fundamental causes of economic prosperity that vary across countries, such as institutions and geography. By looking at regional data within Prussia, all our observations are exposed to the same institutional and legal setting. Similarly, problems of geographical variation are substantially smaller within Prussia than on a global scale, and we control for a rich set of geographical features. We can even include district fixed effects, using only variation across counties within each district. In effect, we hold institutions and geography constant and ask whether there is a role for religion in economic outcomes.[4]

In contrast to the cross-country findings, we do find a significant association between Protestantism and economic prosperity when using county-level variation in late 19th-century Prussia, confirming Weber's descriptive observation. But we also show that there is a strong association between Protestantism and literacy, confirming the basic tenet of our suggestion that Luther's preaching advanced education.[5] When Protestantism and literacy are entered jointly in a "horse race" to explain economic prosperity, the association between Protestantism and economic outcomes vanishes, and the whole effect is absorbed by a significant association between literacy and economic outcomes.[6] Thus, Protestantism does have no independent association with economic prosperity once differences in literacy are accounted for. The same pattern of results holds when we use density of schools rather than literacy as an alternative measure of education, showing that there is a significant role for the supply of schools in the Protestant lead. The interpretation of these findings most favorable to the Weber thesis would be that the economic role of the Protestant ethic works exclusively via human capital accumulation – a thought not explicitly contained in Weber's work.

But we argue below that for several reasons, the higher literacy of Protestant counties is exogenous to ethics, as well as to economic outcomes. Not only was literacy an unintended side effect of Luther's Gospel-reading aims, unrelated to a work ethic or any other economic thought. Also, most of the 19th-

---

[4] In the discussion of "fundamental" causes of economic growth, the cross-country finding by Glaeser, La Porta, Lopez-de-Silanes, and Shleifer (2004) that human capital may be a more basic source of growth than institutions carries particular interest for our study.

[5] In a similar vein, Goldin and Katz (2002) show that in the United States in 1910-1938, areas that led in secondary education had higher shares of Protestant population.

[6] The finding of an important role of education is consistent with a long literature stressing the importance of human capital for historical economic development in general; cf., e.g., Easterlin (1981), Goldin (2001), Lindert (2003), and Galor (2005). Landes (1969) stresses that education was a key ingredient for the transfer of industrial leadership from Britain to Germany at the end of the 19th century. For a recent review of the vast literature on the role of human capital in modern economic growth, cf. Hanushek and Wößmann (2007).

century denominational variation can be traced back to denominational choices of local rulers during the Reformation in 16th and early 17th century, mostly motivated by religious conviction and power politics vis-à-vis the Pope and the German Emperor. Furthermore, Protestantism spread across the German Empire roughly in circles around Luther's city of Wittenberg, whose role as the origin of Protestantism was triggered by a particularly vicious example of indulgence practices.

We use the concentric diffusion of Protestantism to obtain exogenous variation in counties' shares of Protestants in a "double-IV" estimation. Using distance to Wittenberg as an instrument, we can identify an effect of Protestantism on literacy, which is then found to advance economic outcomes. In this model, identification comes from the assumption that Luther's Reformation was an exogenous event, generating a random shock that spread concentrically around Wittenberg. We provide several pieces of evidence showing that the adoption of the Reformation was indeed unrelated to the economic or educational reality of the time.

Combining the facts about the historical origin of the Protestant geographical variation with the results of our county-level estimations, we argue that the variation in literacy used in this paper is exogenous both to any work ethic and to economic outcomes. We also show that literacy had the same effect on economic outcomes in Protestant and Catholic regions, which does not allow for a role of schools in economic progress solely through their potential transmission of a differential work ethic. As a consequence, the higher economic prosperity of Protestant counties has to be viewed as a consequence of higher human capital (unintended during its adoption), but not of work ethic.

Thus, the evidence favors a human capital theory of Protestant economic prosperity over Weber's theory. The driving force of the higher economic prosperity of Protestants in late 19th century Prussia was education. Religion in the sense of an *ethical* disposition towards economic issues did *not* have a significant effect on economic outcomes. Of course, without the Reformation, Protestants would not have acquired more human capital relative to Catholics. In this sense, religion *was* important for economic success, because without intention it involved an uneven accumulation of human capital.

As a final addition, we use individual-level data from contemporary Germany to show that the same pattern between religious denomination, education, and income holds in Germany to these days. The use of contemporary data allows us to go from county-level to individual-level data and to use labor income as an individual-level measure of economic success. Protestants earn more than Catholics even in the late 1990s, but again, this can be fully traced back to their higher level of education.

On a cautionary note, we stress that there is considerable controversy about what Weber's own main hypothesis about Protestantism and the development of capitalism actually was. However, it goes undisputed that the core of his argument is that there is a difference in ethical disposition between